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Expert System (dbase_esystem.php) | Scientists' Profile (drmr_staff.php?emp_cat=Scientific) | Tenders (tender.php) | Proceedings | ICAR-Mail (https://mail.icar.gov.in/) | (https://www.facebook.com/rapeseedmustardresearch/)

(https://twitter.com/icardrmr)

(https://www.youtube.com/channel/UCXjgWUtBObWeasOe2pPI9mQ)

भाकृअनुप
ICAR

(https://www.drmr.res.in/)

ICAR-IIRMR
भाकृअनुप-भारतीय सरसों अनुसंधान संस्थान
ICAR-Indian Institute of Rapeseed-Mustard Research
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Sewar, Bharatpur 321303 (Rajasthan), India



ISO 9001:2015 CERTIFIED

HIGHLIGHTS

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DRMR At A Glance

- About DRMR (index.php)

- Directors (drmr_directors.php)
- From the Director's desk (director_desk.php)
- Organizational setup (organizational_setup.php)
- Cadre strength (cadre_strength.php)
- Research and Admin. units (drmr_units.php)
- DRMR Staff (drmr_staff.php?emp_cat=Scientific)
- Awards and Recognitions (drmr_award.php)

Research & Technology

- Projects at DRMR (drmr_projects.php?project_cat=Institute)
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- (drmr_projects.php?project_cat=Institute)Salient Achievements (drmr_achievements.php?unit_achievement=Crop Improvement)
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- About Rapeseed-Mustard (about_rmcrop.php)
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AICRP-Rapeseed-Mustard

- About AICRP-RM (about_aicrp.php)
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- (aicrp_emp.php?centre_code=PAU&emp_disc=All-Discipline)Performa Entry Evaluation (pdf/proforma_for_entries_evaluation.doc)
- AICRPRM Annual Report 2025 (aicrp2025/AICRP-AR-2025.pdf)
  (aicrp2025/AICRP-AR-2025.pdf)
- (aicrp2025/AICRP-AR-2025.pdf) AICRPRM Proceeding 2025 (aicrp2025/AICRP-Proceeding-2025.pdf)
  (aicrp2025/AICRP-Proceeding-2025.pdf)

Facilities

(aicrp2025/AICRP-Proceeding-2025.pdf)
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- (aicrp2025/AICRP-Proceeding-2025.pdf)Instrument Facility (instruments_facility.php)
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- (instruments_facility.php)Library (drmr_library.php)
- Seed Processing Plants (seed_plant.php)
- Guest House (guest_house.php)
- Guest House Application Form: PDF (pdf/GH-Application.pdf)| MS WORD (pdf/GH-Application.doc)

Technologies Developed

Others

- Mera Gaon Mera Gaurav (MGMG) (mgmg.php)
- (mgmg.php)
- (mgmg.php)Swacch Bharat Mission (swacch_bharat.php)

Crop Improvement

- The first CMS based hybrid (NRCHB 506) and 04 varieties (NRCDR 02, NRCHB 101, NRCDR 601, and DRMRIJ 31) of Indian mustard and one variety of yellow sarson (NRCYS 05-02) were developed at DRMR.
- Fifty novel genetic stocks of rapeseed-mustard (CMS, restorer, low erucic acid, low erucic acid & low glucosinoltes. high oil content, high oleic acid and low linolenic acid, dwarf, earliness, long main shoot,
- bold seed, yellow seed, tetralocular siliquae, white rust resistance, tolerance to high temperature and salinity during juvenile stage, high temperature tolerance during terminal stage and high water use efficiency) including eleven from DRMR were registered with NBPGR, New Delhi till 2013.
- A total of 142 varieties (Indian mustard-91; toria-16; yellow sarson-11; gobhi sarson-11; brown sarson-3; karan rai-4; taramira-5 and black mustard-1) of rapeseed mustard have been released after inception of AICRP-RM in 1967 till 2013. These include six hybrids. Rapeseed-mustard varieties having tolerance to biotic (white rust, Alternaria blight, powdery mildew) & abiotic stresses (salinity, high temperature) and quality traits have been recommended for specific growing conditions.
- 3013.36 q breeder seed was produced under the AICRP-RM from 1984- 85 to 2012-13 against the indent of 1292.98 q.
- Of the 12, 755 accessions available in India, the Directorate have 2,452 accessions (Indian mustard-1868, Toria-245, Yellow sarson-150, Gobhi sarson-105, Taramira-48, Karan rai-17 and Others-19) through acquisition from national /international agencies and collection. Further, a total of 4,580 accessions were distributed to various national/international and non-governmental organizations since establishment.
- Interspecific hybrid derived from cross NRCYS5-02 (B. rapa ssp. yellow sarson) x B. frticulosa (wild species) and Brassica tournefortii x B. rapa var. yellow sarson (NRCYS-05-02), through sexual hybridization was confirmed using morphological, cytological and STMS markers.
- Base populations utilizing promising donors for drought tolerance, high oil content and yield components have been developed. Genetics of important morpho-physiological characters (nodal pigmentation, siliquae orientation, flower colour, water use efficiency, quality traits etc.) has been worked out.
- DUS guidelines for four species of rapeseed mustard and DUS descriptor of 104 varieties were developed.

Last updated: December 07, 2024

Crop Production

- Adoption of reduce tillage practices (3-4 plough/harrow) improves soil health without any loss in mustard seed yield in comparison to conventional tillage (6-10 plough/harrow).
- Furrow irrigated raised bed (FIRB) system improve the mustard seed yield by 10% and saves 33% water over conventional practices.
- Seed dressing with nitrogen fixing bacteria Azotobacter saves up to 30 kg N/ha and increases mustard seed yield up to 21%.
- Seed dressing with phosphorus solubilizing bacteria (PSB) enhance phosphorus use efficiency from 5.2 to 19.8 kg seeds/kg P_2O_5 applied and increases mustard seed yield up to 37% over application of phosphorus only.
- Basal application of 40 kg S/ha through gypsum or elemental sulphur to mustard crop enhances its seed oil content up to 2% and seed yield up to 25%.
- Basal application of 5 kg Zn/ha to mustard crop enhance siliqua/plant, nitrogen use efficiency and thereby seed yield up to 12%.
- Application of boron improves siliqua and seed formation in mustard. Basal application of 1 kg B/ha or 2 foliar spray of 0.25% boric acid at 40 and 60 days after sowing increases mustard seed yield up to 21%.
- Application of 2.5 t/ha vermicompost or FYM + 75% recommended fertility level gives 5-10% higher seed yield over recommended fertilizer (80:40:0 kg/ha).
- Incorporation of 2.5 t/ha mustard residue followed by Sesbania green manuring during kharif enhance mustard seed yield up to 50% over fallow-mustard sequence and improves stable soil organic carbons, infiltration rate, water holding capacity, nutrient retention capacity and soil porosity.

- Integrated application of 2.5t vermicompost or FYM + 80 kg N + 40 kg P2O5 + 40 kg S + 5 kg Zn + 1 kg B per hectare improves the mustard seed yield by more than 10% over recommended fertilizer (80:40:0 kg/ha).
- Use of sprinkler and drip irrigation system improves seed yield of mustard by 25-30% over check basin. It also reduces total requirement of irrigation water by 50% and nitrogen fertilizer by 25%.
- Two hand weedings at 25 and 50 days increases the seed yield by more than 12% over weedy fields. Application of 1.0 kg a.i./ha pendimethalin 30 EC (PE) or 0.06 kg a.i./ha clodinofop 15 WP (at 25-30 DAS) is recommended to controls many annual grasses and some broad leaved weeds in mustard field.
- Adoption of cluster bean-mustard, pearl millet-mustard or Sesbania green manure-mustard sequence improves the total system productivity up to 64% over traditional fallow-mustard practices under protective irrigation.

Crop Protection

- Seed treatment (5g/kg) with talc based formulation of GR isolate of Trichoderma harzianum reduced Sclerotinia and Alternaria diseases up to 30% in Indian mustard.
- Allium sativum bulb extract (1%, w/v) found effective against Alternaria blight and Sclerotinia rot
- Genotypes Bio-YSR, EC-399296, EC 399299, NPJ-127, NRCDR-515, JM-1, EC-399313, JMY-11 were found resistant to white rust.
- New fungal pathogen Nigrospora oryzae was reported to cause stem blight disease in Indian mustard
- Mustard aphid reduced seed yield and oil content up to 86.1% and 3.6%, respectively in B. juncea. The corresponding losses were up to 92.1% and 4.6% in B. napus.
- Total phenol, protein, surface wax, glucosinolates, proline and lectins negatively and total sugars were positively correlated with mustard aphid infestation.
- Neem oil (2%) and neem seed kernel extract (5%) reduced mustard aphid infestation by 80%.
- Verticillium lecanii @ 108 CS/ml reduced aphid population by 80% under humid conditions.
- Two new insect pests i.e. Gonocephalum sp. (Coleoptera: Tenebrionidae) and Tanymecus indicus Fst. (Coleoptera: Curculionidae) were recorded during seedling stage.
- Seed treatment (ST) carbendazim 50WP 2g/Kg seed+no irrigation during 25th December to 15th January+ foliar spray (carbendazim 50WP @ 0.2%) at 45-50 and 65-70 DAS for effective management of stem rot.
- Integrated disease management module developed with seed treatment T. harzianum @10g/kg seed, soil application of Trichoderma (1 kg/ 50kg FYM), Basal application of zinc sulphate @ 15 kg/ ha + S (dose location specific) + Boron (10 kg borax/ ha), line sowing 45X20 cm (RxP), no irrigation during 25th Dec to 15th Jan.
- Seed treatment with Trichoderma harzianum @ 10 g/kg seed followed by foliar spray of Pseudomonas fluorescence (oil-based) @ 10ml/1 at flower initiation stage for reducing the disease (white rust, Sclerotinia rot, downy mildew and Alternaria blight).
- Seed treatment with metalaxyl + Foliar spray of mancozeb 0.2% 45 DAS followed by metalaxyl + mancozeb 0.2% 60 DAS proved effective in controlling the white rust.
- Foliar spray of propiconazole 25 EC @ 0.05% was proved effective against powdery mildew and Sclerotinia rot.
- Germplasm DRMR-2019 (IC0598622; INGR17077) and DRMR-2035 (IC0598623; INGR17078) of Indian mustard (B. juncea) was registered for white rust resistance by PGRC, New Delhi.
- Germplasm, RH 1222-28, EC 597328, EC 766553, EC 766620, EC 765048, IC 492687, IC 492690, IC 492695 and IC 511651 were reported tolerant to Sclerotinia stem rot (lesion size <3.0 cm and disease incidence<10%).
- Development of non-injury inoculation technique for assessing Sclerotinia stem rot in oilseed Brassica.
- Seed treatment with imidacloprid 70 WS @ 5g/kg of seed provided good control to painted bug up to 30-35 DAS.
- Spray of dimethoate @ 1 ml/litre followed by release of Coccinella septempunctata @ 5000 beetles/ha is recommended for the eco-friendly integrated pest management of mustard aphid.
- Foliar spray with NSKE @ 5% or neem oil @2% for the management of mustard aphid as an alternative to chemical control

Biochemistry & Biotechnology

- Near Infra Red (NIR) was standardized for simultaneous determination of percent moisture, oil and protein content in intact seed.
- Palladium complex method developed for estimation of total glucosinolate content in seed meal of rapeseed-mustard using ELISA READER.
- Soxhlet, NMR and NIR methods for oil estimation in rapeseed-mustard compared.
- FT-NIRS calibrations developed for determination of oil, protein, fatty acid and total glucosinolate content in intact seed of rapeseed-mustard.
- Glucosinolate profiling of thirty seven rapeseed-mustard cultivars using HPLC was carried out.
- Variation of total glucosinolate in different plant parts of mustard cultivars (both high and low) at three different developmental stages recorded.
- Antioxidant activities of extract of rapeseed-mustard leaves and seed meal carried out using DPPH, ABTS and FRAP assay.
- 104208, 103208, 24658, 9020 and 400 samples of rapeseed-mustard were analyzed for oil, protein, glucosinolate, fatty acid and crude fibre content, respectively.
- A high frequency regeneration protocol for Brassica juncea cv. NRCDR-2 has been standardised using cotyledonary petiole explants. The best result was obtained on MS medium supplemented with 1.0 mg/l BAP and 0.10 mg/l NAA, where it gave an average no. of 4.56 shoots per explants and 86.66% explants regeneration. The highest percentage of root regeneration (100%) was obtained on MS medium supplemented with 0.30 mg/l IAA.
- Twenty one different wild species belonging to ten different genera of Brassicaceae are being maintained and genomic STMS markers suitable for their molecular characterization have been identified.

Technology Assessment and Dissemination Division

- The unit has forged constructive and fruitful linkages with several institutions In its endeavour to bring the scientific technologies to the farmers. These linkages have lent wider reach and effectiveness to the transfer of technology programmes.
- Adoption practice of 12 major cultivation practices across the major rapeseed-mustard growing regions of Rajasthan, Haryana and Uttar Pradesh were quantified based on an adoption index which indicated low adoption of seed treatment, pest management and soil treatment with mean percentile scores of 23.2, 16.3, and 5.4 respectively. The study also identified 5 economic, 8 infrastructural, 4 agro-climatic and 5 technological constraints affecting the adoption of mustard technology.
- A study of 5 districts of Bharatpur division was done to identify the pattern of adoption and reasons of non-adoption of technology based on key informant responses. The lack of knowledge and conviction, lack of soil moisture, reduction in soil fertility, problem of saline water, non-availability of fertilizers etc. were the reasons for gap in adoption of fertilizer recommendation.
- Since the turn of this century, the unit has conducted more than 350 Frontline demonstrations More than 10,000 farmers have benefitted from scientist-farmer interaction meetings through kisan ghostis, sarson farm school, farmer Interest groups, and village meetings.
- The visitor advisory services have benefitted more than 50,000 farmers. The unit has organized more than 75 training programmes for the benefit of scientist, extension personnel, farmers and line department officials. More than 600 sponsored radio programmes and 50 TV stories have been broadcasted and 200 tribal farmers have received training in rapeseed-mustard production technology.
- Participatory varietal selection and evaluation (PSVE) done at the directorate with 45 farmers of 3 villages in Bharatpur. Farmers evaluated 13 Indian mustard varieties for seed yield and other characters. On the basis of scoring, NRCDR-02, NRCHB-101, RB-50, NRCHB-506 and Geeta were most preferred by the participating farmers.
- The efforts of the unit has transformed Unchagaon village of Bharatpur district in Rajasthan, which was adopted by DRMR, through its catalytic role in bringing together various government agencies and other stakeholders together to address critical developmental challenges in the village and through intensive extension efforts to popularize of improved rapeseed-mustard production technology. The mustard productivity in the village increased by nearly 40 per cent due to the technology intervention efforts

RESOURCES

- Germplasm Indent & MTA ()
- Rapeseed-Mustard Quality Seed ()
- DRMR Logo ()
- DRMR Silver Jubilee Logo ()

SERVICES & FACILITIES

- Automatic seed processing ()
- Organizing seed fortnight ()
- Guest house ()
- Library ()

IMPORTANT LINKS

- ICAR-Unified Communications ()
- FMS/MIS of ICAR ()
- HYPM ()

E-GOVERNANCE















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